

Automated Detection of IPL Teams

Using Jersey Identification

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□ The Problem

- Input: IPL broadcast image (800×600 px)
- Divide into 8×8 grid = 64 cells (100×75 px each)
- Predict each cell: 0 (none) or 1–10 (IPL team)
- Constraint: NO CNNs — hand-crafted features only
- Output: CSV with columns c01...c64, values 0–10

3,605

Images

194

Features

□ What We Built

- 3,605 labelled images, 44 match scenarios
- 194 hand-crafted features per cell
- Two-stage LightGBM pipeline
- Spatial context classifier (+0.030 F1)
- None-class capping (3× team cells)
- Player detection via connected components

0.712

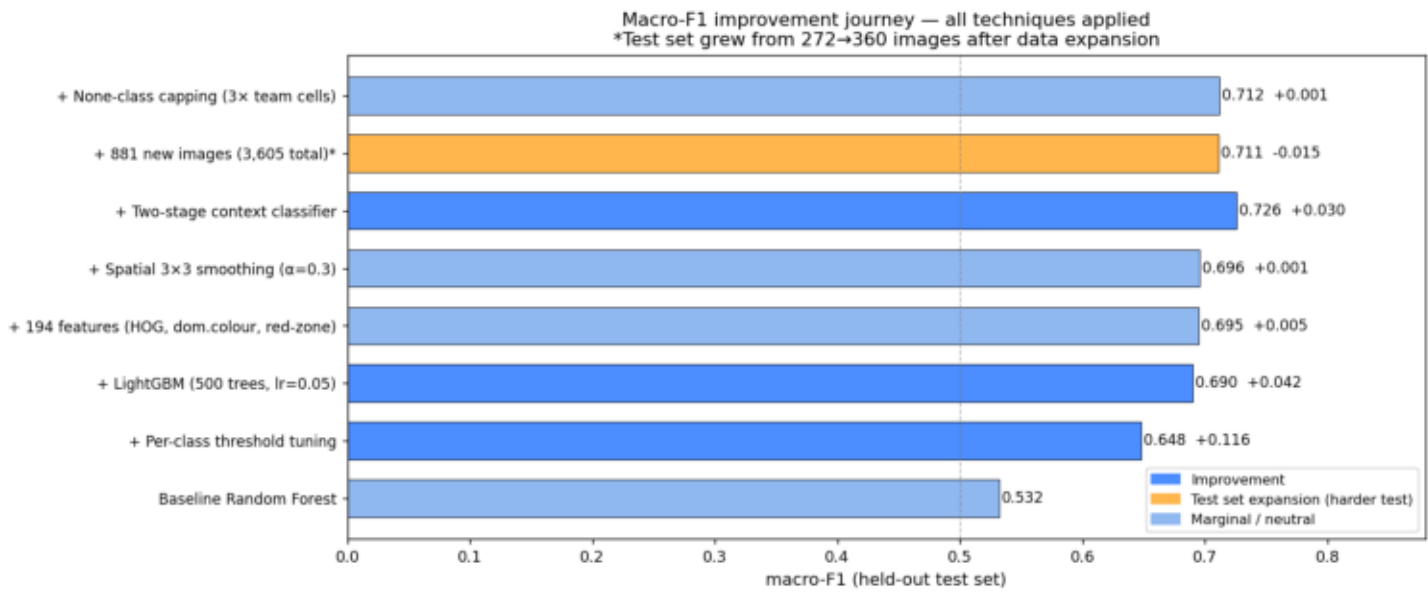
Macro-F1

44

Match Scenarios

Metric	Value
Dataset	3,605 images 44 matches 10 IPL teams 230,720 labelled cells
Features	194 hand-crafted (HSV, HOG, LBP, texture, colour) — No CNN
Best model	Two-stage LightGBM (Stage-1: 194 features → Stage-2: 22 context features)
Macro-F1 (test)	0.712 on 360 held-out images (never seen during training)
Class imbalance	3-layer fix: class_weight + threshold tuning + none-class capping
Hardest case	PBKS vs RCB (both red jerseys) — correctly separated using red-zone histogram
Player detection	Connected-component analysis on cell grid — classical, no CNN

📌 Key Insight:



Step	Technique	F1	Gain
1	Random Forest	0.532	—
2	+ Threshold tuning	0.648	+0.116
3	+ LightGBM	0.690	+0.042
4	+ 194 features	0.695	+0.005
5	+ Spatial smooth	0.696	+0.001
6	+ Stage-2 context	0.726	+0.030
7	+ 881 images*	0.711	—
8	+ None cap	0.712	+0.001

*Test set grew 272→360 images; recall improved on 7/10 classes

The Task

- Input: 800×600 px IPL broadcast image
- Divide into 8×8 grid → 64 cells (100×75 px each)
- Classify each cell: 0=none or 1–10 (IPL franchise)
- Output: CSV — Image File Name, Train Or Test, c01...c64

Label scheme:

⚠ Why Hard

- 83.8% cells = background
- Severe imbalance
- PBKS + RCB both red
- Colour ambiguity
- Variable camera distance
- Cell coverage varies

Dataset Stats

Images: 3,605

Match scenarios: 44

Labelled cells: 230,720

Spec: 800×600 px, 4:3

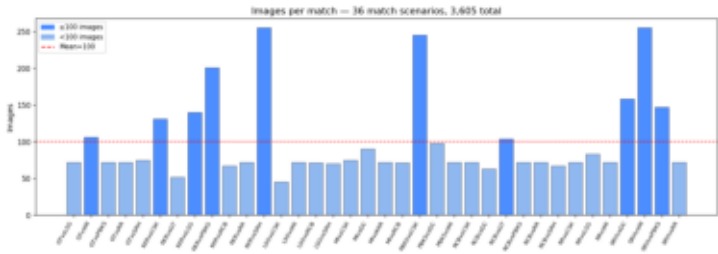
Season: IPL 2025 only

Image Sources

- iplt20.com — official 2025 galleries
- BCCI official media releases
- ESPNcricinfo / Cricbuzz

Labelling Process

- Download 800×600 images
- Apply 8×8 grid overlay (tool)
- Label each cell 0–10
- Export via class tool
- Merge all groups → master CSV
- Validate & resolve conflicts



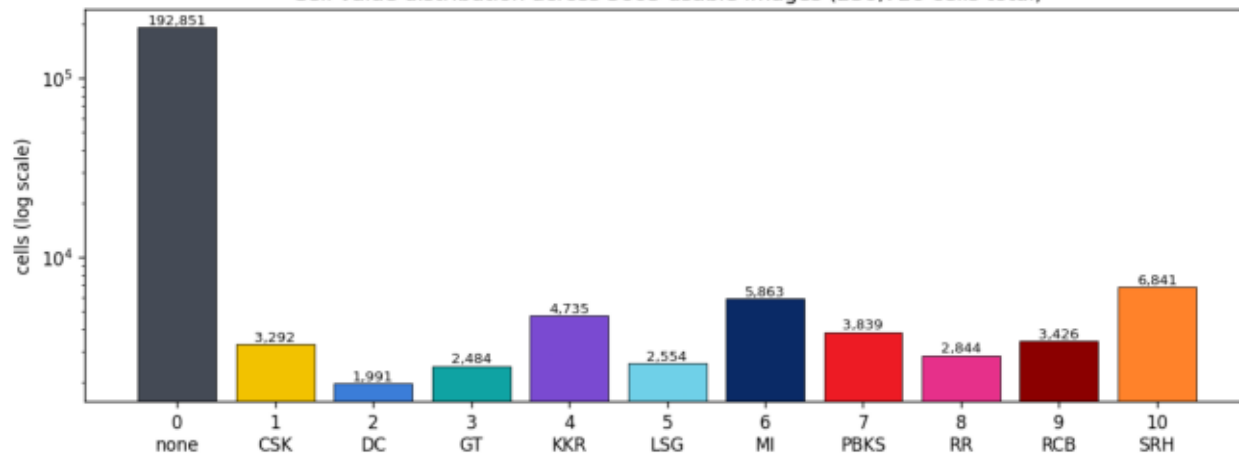
Key Finding

DC: 262 images — fewest → lowest F1 (0.61)

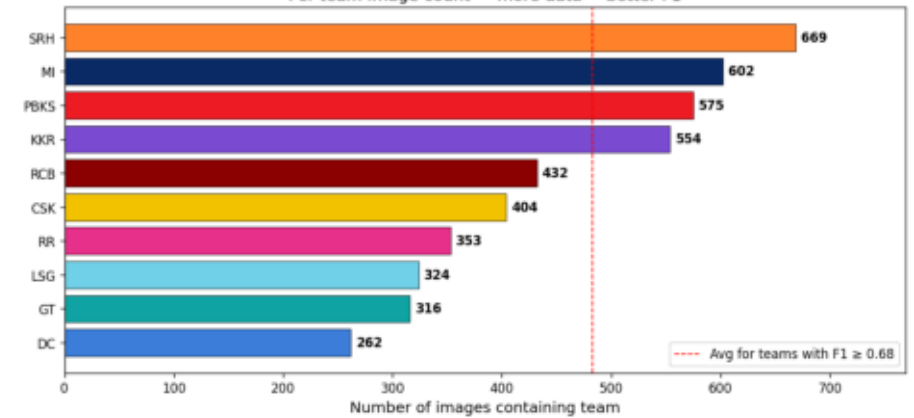
Added 881 new images mid-project

KKR: +278% cells

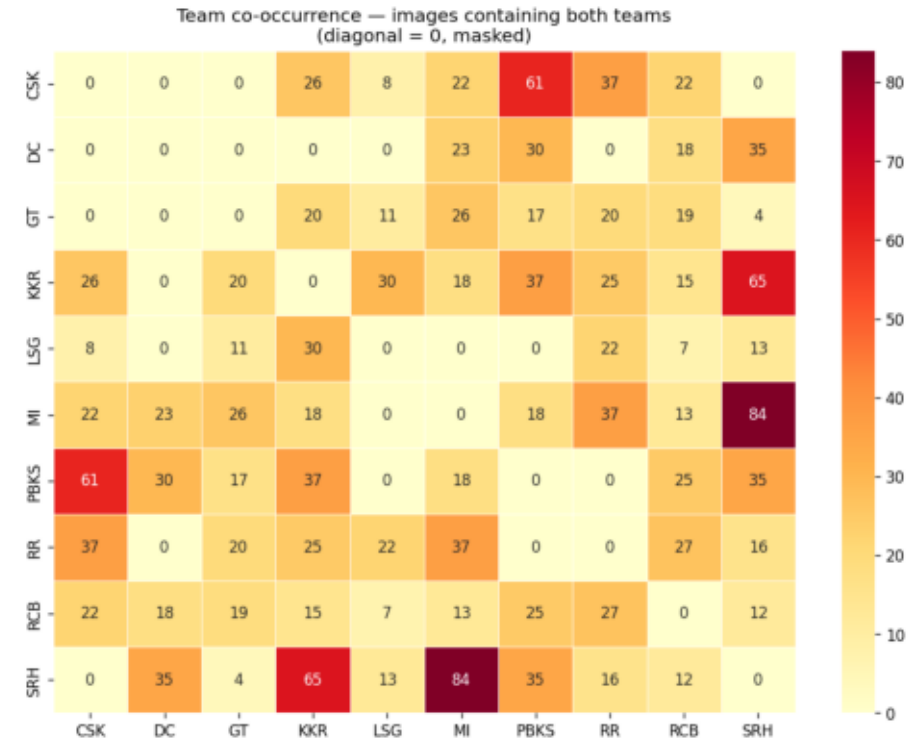
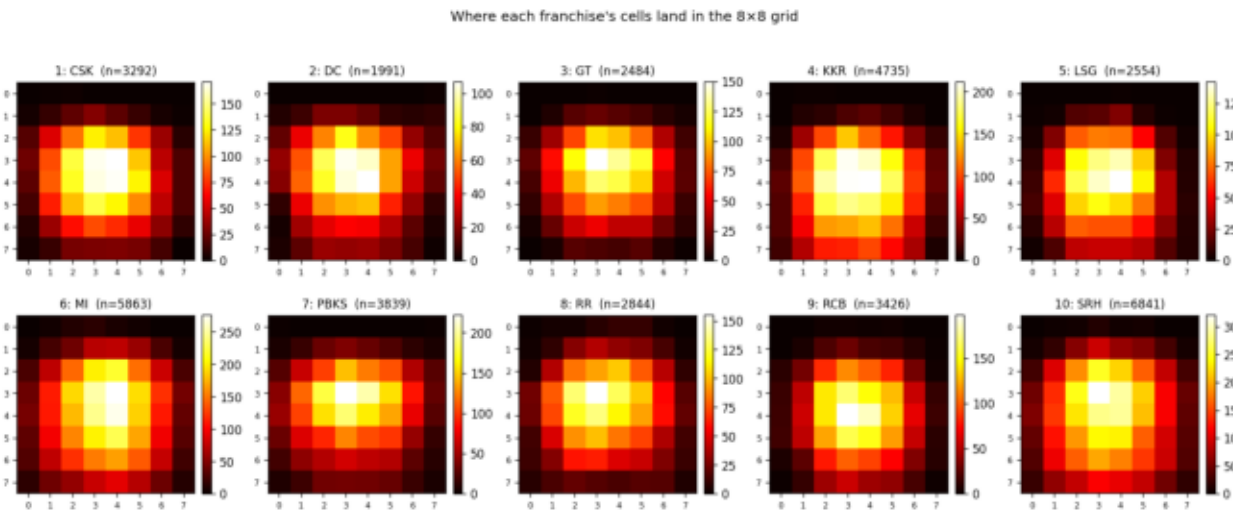
Cell-value distribution across 3605 usable images (230,720 cells total)



Per-team image count — more data = better F1



□ 83.8% of cells are "none" — fixed with class_weight + threshold tuning + none-cap □ Data quantity predicts F1: DC (262 imgs)→0.61, SRH (669)→0.76

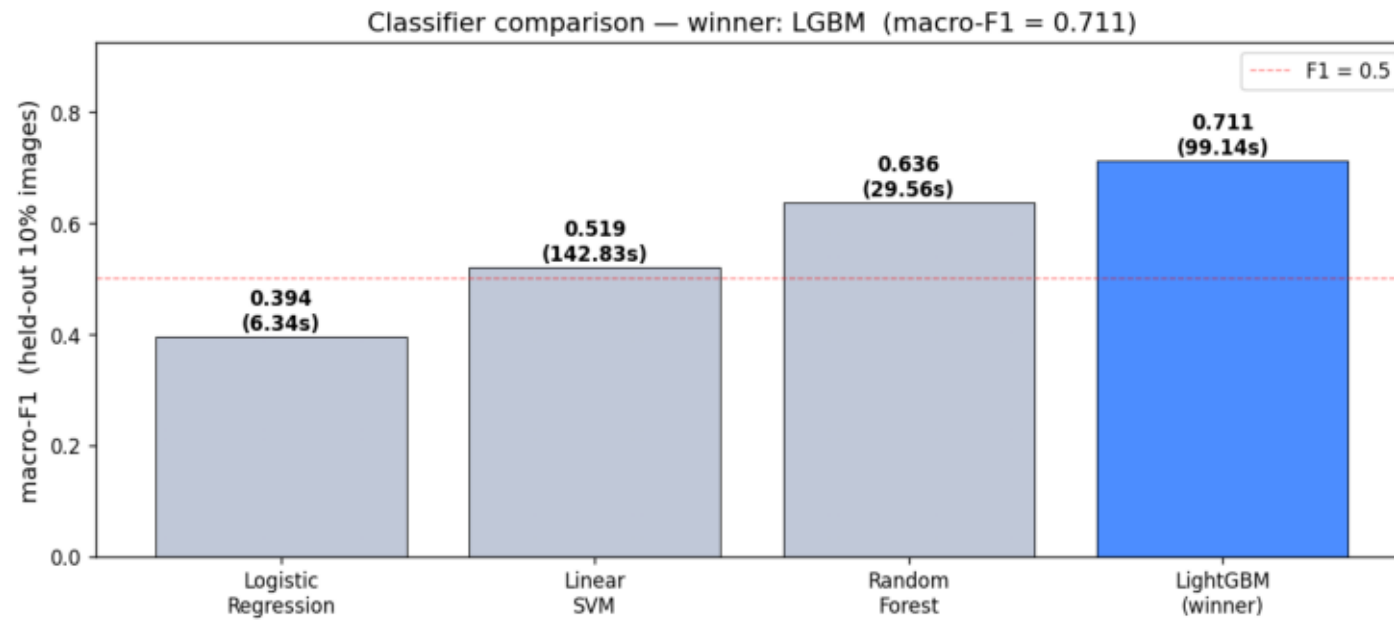


□ Players in rows 2-6 consistently → position features added to 194-D vector

□ Teams only co-occur in actual match fixtures — dataset is match-realistic

Feature Block	Dim	What It Captures	Key For
HSV histograms (H+S+V)	48	Team colour signature	All teams
Mean + Std BGR	6	Average colour tone	Baseline
Position (row, col)	2	Spatial prior — logos in fixed zones	All teams
LBP texture (P=8,R=1)	10	Jersey weave vs smooth background	Texture
Canny edge density	1	Logo boundaries, text overlays	Logos
HOG ← Key addition	96	Jersey stripe / texture direction	All teams
Secondary colour fractions	5	silver(PBKS), black(RCB), gold	PBKS / RCB
Dominant colour pair (k=2)	8	Top-2 colours + proportion	Red teams
Red-zone histogram	16	Fine hue in [0°-25°] + [155°-180°]	PBKS vs RCB
Checkered pattern	2	PBKS distinctive check pattern	PBKS
TOTAL	194		

Why LightGBM Wins



- Gradient boosting handles

class imbalance well

- Native class_weight support

- No scaling needed (unlike SVM)

- Leaf-wise tree growth

→ more efficient than RF

① `class_weight="balanced"`

Adjusts loss function weights

All classes contribute equally

Model penalised equally for

missing PBKS vs none cell

Always active — baseline fix

② Per-class Threshold Tuning +0.116

`argmax(proba / threshold)`

Lower `threshold[c]` → more

willing to predict class `c`

Tuned via coordinate ascent

on calibration split (no leak)

③ None-class Capping ← NEW

Cap none at 3× team cells

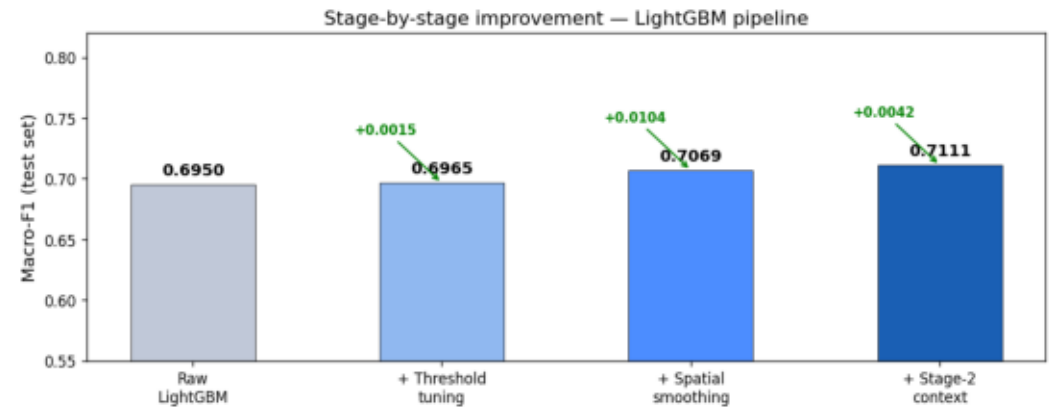
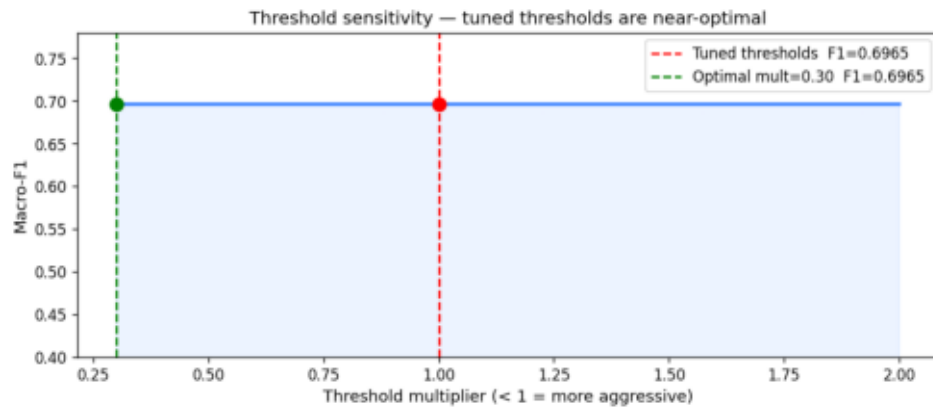
139,010 → 81,402 none cells

(41% reduction)

Reduces background pattern

memorisation → forces model

Combined: `class_weight` corrects loss | threshold tuning adjusts boundary | capping reduces memorisation



Architecture

Stage-1: LightGBM on 194 features

→ (64, 11) probability map

Context features per cell:

[own_proba(11) | neighbour_mean_proba(11)]

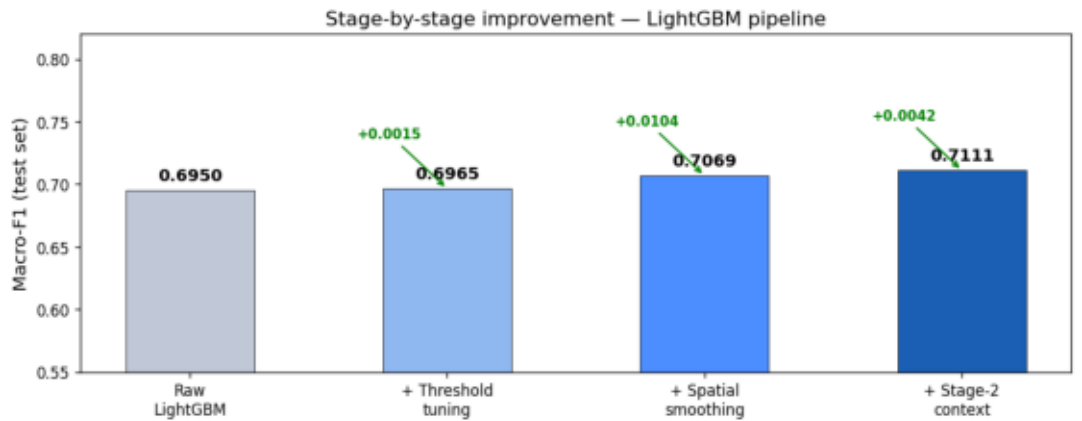
= 22 features total

No Data Leakage

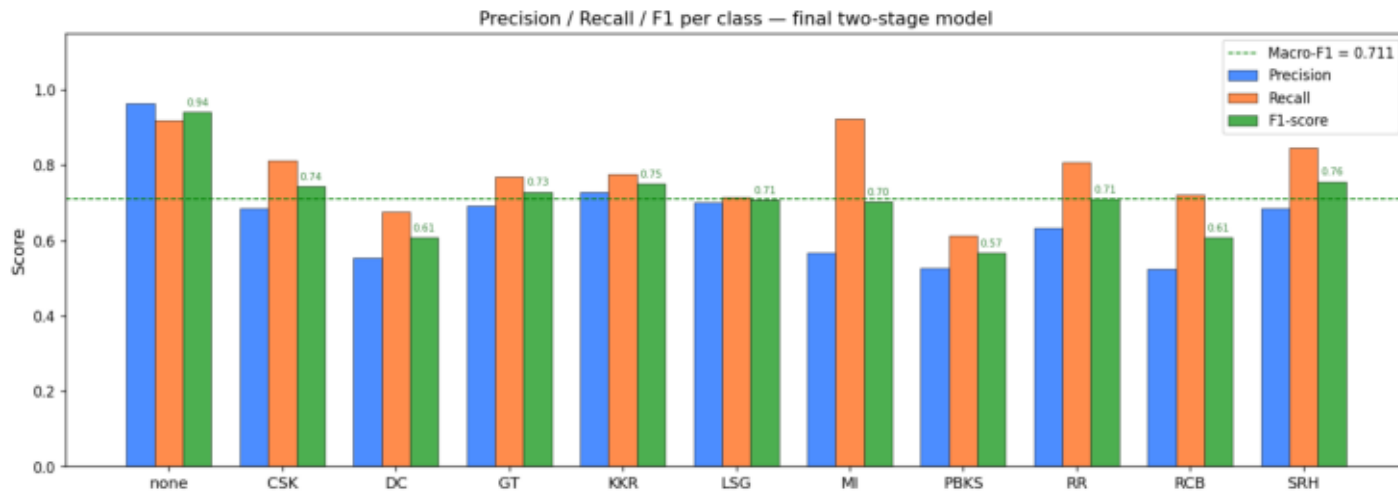
Stage-1 trained on fit cells (80% of train)

Stage-2 trained on cal cells (20%) — out-of-sample for Stage-1

Evaluated on test cells (10%) — never touched by either stage



Class	Recall Before	After	Δ
MI	0.82	0.92	+0.10 ↑
DC	0.62	0.68	+0.06 ↑
GT	0.72	0.77	+0.05 ↑
PBKS	0.57	0.61	+0.04 ↑
RCB	0.70	0.72	+0.02 ↑



Key Observations

□ Precision strong (0.52–0.96)

across all teams

Confusion matrix — counts (23,040 test cells)

True \ Predicted	none	CSK	DC	GT	KKR	LSG	MI	PBKS	RR	RCB	SRH
none	17710	144	108	69	142	84	358	177	101	148	258
CSK	76	336	0	0	1	0	1	0	0	0	0
DC	56	0	136	0	2	1	6	0	0	0	0
GT	46	0	0	160	0	0	0	0	0	2	0
KKR	93	2	0	0	401	0	17	0	0	0	4
LSG	69	0	0	1	0	208	6	0	0	7	0
MI	42	0	1	0	0	0	519	0	0	0	0
PBKS	104	2	0	1	0	0	0	222	1	27	5
RR	35	0	0	0	4	0	3	0	176	0	0
RCB	52	1	1	0	0	4	0	21	0	204	0
SRH	92	5	0	0	1	0	5	2	0	1	579

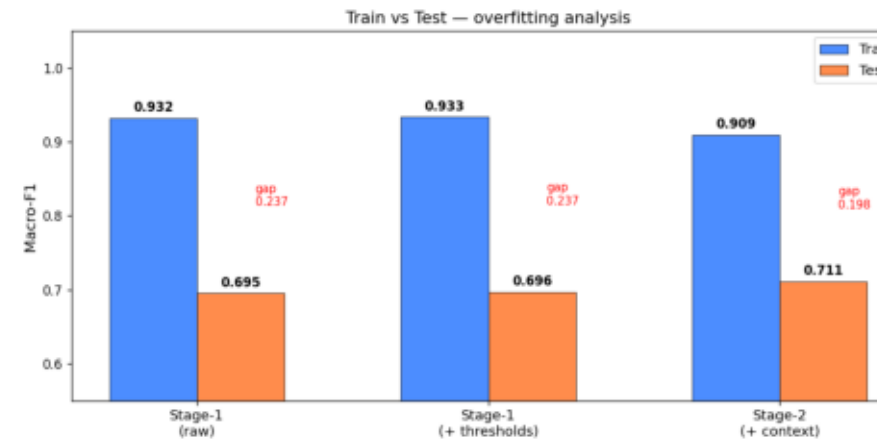
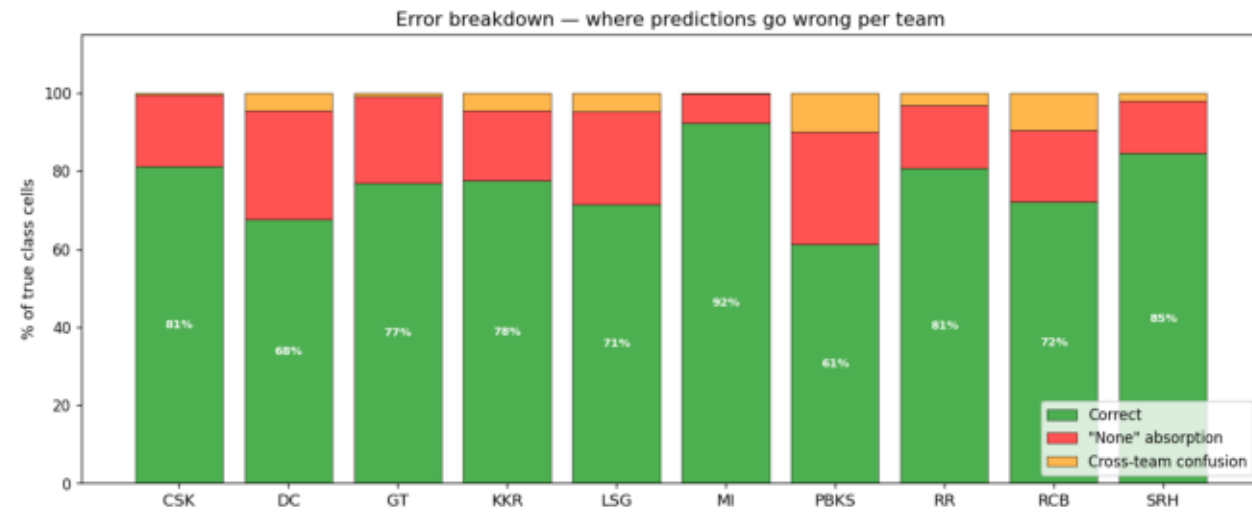
Row-normalised (recall per class)

True \ Predicted	none	CSK	DC	GT	KKR	LSG	MI	PBKS	RR	RCB	SRH
none	0.92	0.01	0.01	0.00	0.01	0.00	0.02	0.01	0.01	0.01	0.01
CSK	0.18	0.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
DC	0.28	0.00	0.65	0.00	0.01	0.00	0.03	0.00	0.00	0.00	0.00
GT	0.22	0.00	0.00	0.77	0.00	0.00	0.00	0.00	0.00	0.01	0.00
KKR	0.18	0.00	0.00	0.00	0.78	0.00	0.03	0.00	0.00	0.00	0.01
LSG	0.24	0.00	0.00	0.00	0.00	0.71	0.02	0.00	0.00	0.02	0.00
MI	0.07	0.00	0.00	0.00	0.00	0.00	0.92	0.00	0.00	0.00	0.00
PBKS	0.29	0.01	0.00	0.00	0.00	0.00	0.00	0.61	0.00	0.07	0.01
RR	0.16	0.00	0.00	0.00	0.02	0.00	0.01	0.00	0.81	0.00	0.00
RCB	0.18	0.00	0.00	0.00	0.00	0.01	0.00	0.07	0.00	0.72	0.00
SRH	0.13	0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.85

△ Recall is the weakness

Model is conservative —

misses cells vs wrong preds



□ #1 failure: "None" absorption (23-41% of team cells → background)

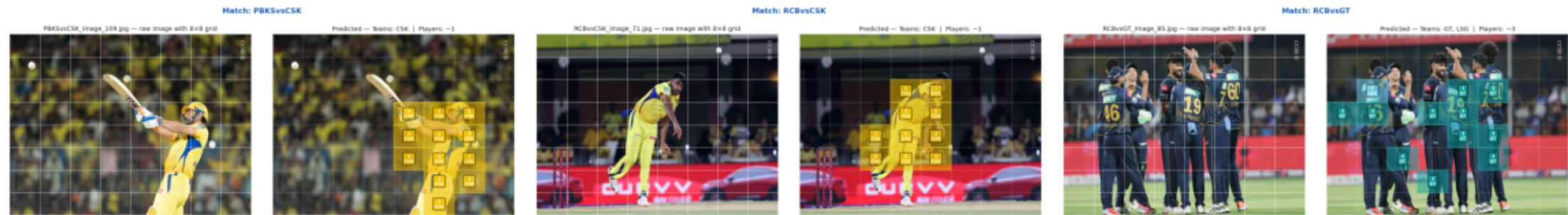
Train-Test gap ~0.23

"Our test macro-F1 of 0.712 is the valid metric. The 0.94 training score reflects tree memorisation of specific backgrounds — not poor generalisation. The gap is expected and NOT harmful — regularisation attempt confirmed this."

Left: Raw image + 8x8 grid overlay Right: Predicted team cells highlighted



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Two-Method Pipeline

1. HOG person detector (cv2)

→ pixel bounding boxes

→ fails in stadium crowds

2. Connected-component analysis

per-team cell blobs

Player Count Failure Modes

- Same-team merge: 4×MI → 3 (adjacent blobs fuse)
- Non-upright pose: 5 players → 3 (lying/bent → few cells)
- Extreme close-up: 1×RR → 2 (fills whole frame)
- Wide-angle: 8 players → 4 (<6 cells each at distance)

Fundamental limit: at 8×8 resolution, a group of upright

Overall (demo images)	Correctly detected
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Approach	What We Tried	Why It Failed	Lesson Learned
Gabor filters	4 angles×2 freq per cell	74ms/image, zero F1 gain	HOG already captures texture
PCA reduction	50–150 components	Every level hurt F1 (−0.07)	LightGBM selects features itself
CatBoost	Discussed as alternative	All numerical features → LightGBM wins	Match tool to data type
HOG for counting	cv2 pedestrian detector	9 false positives in stadium crowds	HOG trained on streets not cricket
PCA pose estimation	Blob orientation → pose	Group of players = horizontal blob	8×8 too coarse for pose
Pose-adjusted divisors	÷9/÷7/÷5 by pose class	Score 6/13 vs fixed ÷12 → 7/13	Simpler beats complex here
Regularised LightGBM	Fewer trees + L1/L2	Test F1 dropped 0.711→0.702	Model learned real signal

⚠ Challenges

1. PBKS vs RCB red-vs-red confusion

Both wear red. Required 3 targeted features.

Still 6–8% cross-team confusion remains.

2. 83.8% class imbalance

3-layer fix needed. "None absorption"

📌 Key Learnings

1. Threshold tuning > feature engineering

Biggest gain (+0.116) came from tuning

decision boundaries, not adding features.

2. Spatial context matters most

Stage-2 (+0.030) outperformed all

□ What Works

- Team classification: macro-F1 = 0.712

on 360 completely unseen images

- Correctly separates PBKS vs RCB (both red)
- Full pipeline: one .pkl → predictions CSV
- 194 hand-crafted features, zero CNN
- 3-layer imbalance fix: class_weight +

⚠ Known Limits

- PBKS (0.57), DC (0.61) — data poverty
- Player counting: ~58% — 8×8 resolution limit
- Train-test gap: 0.23 — tree memorisation

□ Future Work

- 200 more DC match images

→ DC F1: 0.61 → ~0.70

- Label 447 remaining images

→ PBKS + RCB improvement